

librispeech_validated_all_voting_ensemble

scalars_librispeech_validated_all_voting_ensemble

key	value
n_samples	2611
main_kwargs_applied	{}
timings_seconds.cross_wer	1.5
timings_seconds.ranking	82.3
timings_seconds.model_filtration	518.4
timings_seconds.rank_distribution	683.5
ranking_stats.n_models	7
ranking_stats.pool	seamless_m4t+whisper_v3+canary+conformer+parakeet+sb+cohere
rank_distribution_summary.n_subpools	64
rank_distribution_summary.n_boot_inner	1000
rank_distribution_summary.n_fusions	99
rank_distribution_summary.n_scorings	294
rank_distribution_summary.seconds_fuse_and_score	160.7
rank_distribution_summary.seconds_bootstrap	4.4
rank_distribution_summary.weighting	filtration
rank_distribution_summary.exponent	1
rank_distribution_summary.mean_entropy	0.3362
rank_distribution_summary.max_entropy	0.3811
rank_distribution_summary.n_models_modal_differs_from_full	0

ranking_librispeech_validated_all_voting_ensemble

model	role	n_samples	wer_pseudo_loo	cer_pseudo_loo	rank_pseudo
parakeet	voter	2611	1.0123	0.3066	1
cohere	voter	2611	1.0184	0.3334	2
canary	voter	2611	1.2738	0.4301	3
conformer	voter	2611	1.3373	0.3973	4
sb	voter	2611	1.5533	0.4847	5
seamless_m4t	voter	2611	1.874	0.7039	6
whisper_v3	voter	2611	1.9846	1.1854	7

model_confidence_librispeech_validated_all_voting_ensemble

model	rank_pseudo_full	rank_modal	p_modal	p_at_full_rank	entropy_norm	rank_mean	rank_sd	rank_ci_lo	rank_ci_hi	agrees_with_full
parakeet	1	1	0.5863	0.5863	0.3769	1.4259	0.5197	1	2	True
cohere	2	2	0.5751	0.5751	0.3811	1.6021	0.5163	1	2	True
canary	3	3	0.7093	0.7093	0.3633	3.2442	0.4961	3	4	True
conformer	4	4	0.7067	0.7067	0.3611	3.7567	0.489	3	5	True
sb	5	5	0.9326	0.9326	0.1551	5.0131	0.2716	4	6	True
seamless_m4t	6	6	0.6795	0.6795	0.3413	6.3048	0.477	6	7	True
whisper_v3	7	7	0.6855	0.6855	0.3744	6.6532	0.5408	5	7	True

rank_distribution_librispeech_validated_all_voting_ensemble

model	rank_1	rank_2	rank_3	rank_4	rank_5	rank_6	rank_7
seamless_m4t	0	0	0	0	0.0078	0.6795	0.3126
whisper_v3	0	0	1.3487e-05	0.0006	0.0311	0.2827	0.6855
canary	0.0023	0.0212	0.7093	0.2646	0.0027	0	0
conformer	2.7208e-05	0.0015	0.266	0.7067	0.0257	6.0889e-05	0
parakeet	0.5863	0.4022	0.0109	0.0006	0	0	0
sb	0	0	0.0003	0.0275	0.9326	0.0377	0.0018
cohere	0.4114	0.5751	0.0135	0	0	0	0

pair_probability_librispeech_validated_all_voting_ensemble

pair	pred_winner_full	gap_wer_pseudo	p_first_beats_second	p_supports_full
parakeet cohere	parakeet	0.0061	0.5862	0.5862
seamless_m4t whisper_v3	seamless_m4t	0.1106	0.6858	0.6858
canary conformer	canary	0.0634	0.7321	0.7321
whisper_v3 sb	sb	0.4313	0.0322	0.9678
conformer sb	conformer	0.216	0.9746	0.9746
canary cohere	cohere	0.2555	0.0146	0.9854
canary parakeet	parakeet	0.2616	0.0114	0.9886
seamless_m4t sb	sb	0.3207	0.0092	0.9908
canary sb	canary	0.2795	0.9971	0.9971
conformer parakeet	parakeet	0.325	0.0017	0.9983
whisper_v3 conformer	conformer	0.6473	0.0005	0.9995
conformer cohere	cohere	0.3189	0.0002	0.9998
whisper_v3 canary	canary	0.7107	9.4437e-05	0.9999
parakeet sb	parakeet	0.541	1	1
whisper_v3 cohere	cohere	0.9662	0	1
seamless_m4t cohere	cohere	0.8556	0	1
seamless_m4t parakeet	parakeet	0.8617	0	1

pair	pred_winner_full	gap_wer_pseudo	p_first_beats_second	p_supports_full
seamless_m4t conformer	conformer	0.5367	0	1
seamless_m4t canary	canary	0.6002	0	1
whisper_v3 parakeet	parakeet	0.9723	0	1
sb cohere	cohere	0.5349	0	1

model_filtration_librispeech_validated_all_voting_ensemble

model	n_subpools	f_bar	f_sd	f_min	f_max	survival
parakeet	42	0.0561	0.0101	0.0379	0.077	0.9439
cohere	42	0.0577	0.0088	0.0448	0.0781	0.9423
canary	42	0.087	0.0126	0.0628	0.1126	0.913
whisper_v3	42	0.1014	0.0105	0.0812	0.126	0.8986
conformer	42	0.1078	0.0122	0.0862	0.1371	0.8922
sb	42	0.1174	0.0132	0.0942	0.1471	0.8826
seamless_m4t	42	0.1647	0.0143	0.1314	0.1881	0.8353

by_pool_size_librispeech_validated_all_voting_ensemble

k	pools	total_weight	mean_entropy	max_entropy
4	35	0.4762	0.2317	0.3629
5	21	0.3571	0.2246	0.2722
6	7	0.1429	0.247	0.2988
7	1	0.0238	0.2672	0.2672

cross_wer_librispeech_validated_all_voting_ensemble

	seamless_m4t	whisper_v3	canary	conformer	parakeet	sb	cohere
seamless_m4t	0	2.84	2.21	2.38	2.14	2.56	2.14
whisper_v3	2.84	0	2.35	2.6	2.22	2.81	2.16
canary	2.21	2.35	0	1.9	1.27	2.16	1.54
conformer	2.38	2.6	1.9	0	1.6	1.89	1.6
parakeet	2.14	2.22	1.27	1.6	0	1.88	1.29
sb	2.56	2.81	2.16	1.89	1.88	0	1.74
cohere	2.14	2.16	1.54	1.6	1.29	1.74	0

subpool_filtration_librispeech_validated_all_voting_ensemble (first 60 of 294 rows)

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+canary+conformer	4	seamless_m4t	2611	354	0.1356
seamless_m4t+whisper_v3+canary+conformer	4	whisper_v3	2611	222	0.085
seamless_m4t+whisper_v3+canary+conformer	4	canary	2611	195	0.0747
seamless_m4t+whisper_v3+canary+conformer	4	conformer	2611	280	0.1072
seamless_m4t+whisper_v3+canary+parakeet	4	seamless_m4t	2611	408	0.1563
seamless_m4t+whisper_v3+canary+parakeet	4	whisper_v3	2611	251	0.0961
seamless_m4t+whisper_v3+canary+parakeet	4	canary	2611	172	0.0659
seamless_m4t+whisper_v3+canary+parakeet	4	parakeet	2611	134	0.0513
seamless_m4t+whisper_v3+canary+sb	4	seamless_m4t	2611	343	0.1314
seamless_m4t+whisper_v3+canary+sb	4	whisper_v3	2611	212	0.0812
seamless_m4t+whisper_v3+canary+sb	4	canary	2611	195	0.0747
seamless_m4t+whisper_v3+canary+sb	4	sb	2611	293	0.1122
seamless_m4t+whisper_v3+canary+cohere	4	seamless_m4t	2611	389	0.149
seamless_m4t+whisper_v3+canary+cohere	4	whisper_v3	2611	223	0.0854
seamless_m4t+whisper_v3+canary+cohere	4	canary	2611	212	0.0812
seamless_m4t+whisper_v3+canary+cohere	4	cohere	2611	146	0.0559
seamless_m4t+whisper_v3+conformer+parakeet	4	seamless_m4t	2611	388	0.1486
seamless_m4t+whisper_v3+conformer+parakeet	4	whisper_v3	2611	237	0.0908
seamless_m4t+whisper_v3+conformer+parakeet	4	conformer	2611	260	0.0996
seamless_m4t+whisper_v3+conformer+parakeet	4	parakeet	2611	128	0.049
seamless_m4t+whisper_v3+conformer+sb	4	seamless_m4t	2611	369	0.1413
seamless_m4t+whisper_v3+conformer+sb	4	whisper_v3	2611	244	0.0935
seamless_m4t+whisper_v3+conformer+sb	4	conformer	2611	229	0.0877
seamless_m4t+whisper_v3+conformer+sb	4	sb	2611	247	0.0946
seamless_m4t+whisper_v3+conformer+cohere	4	seamless_m4t	2611	398	0.1524
seamless_m4t+whisper_v3+conformer+cohere	4	whisper_v3	2611	221	0.0846
seamless_m4t+whisper_v3+conformer+cohere	4	conformer	2611	268	0.1026
seamless_m4t+whisper_v3+conformer+cohere	4	cohere	2611	123	0.0471
seamless_m4t+whisper_v3+parakeet+sb	4	seamless_m4t	2611	376	0.144
seamless_m4t+whisper_v3+parakeet+sb	4	whisper_v3	2611	231	0.0885
seamless_m4t+whisper_v3+parakeet+sb	4	parakeet	2611	147	0.0563
seamless_m4t+whisper_v3+parakeet+sb	4	sb	2611	293	0.1122
seamless_m4t+whisper_v3+parakeet+cohere	4	seamless_m4t	2611	428	0.1639
seamless_m4t+whisper_v3+parakeet+cohere	4	whisper_v3	2611	234	0.0896
seamless_m4t+whisper_v3+parakeet+cohere	4	parakeet	2611	159	0.0609
seamless_m4t+whisper_v3+parakeet+cohere	4	cohere	2611	142	0.0544
seamless_m4t+whisper_v3+sb+cohere	4	seamless_m4t	2611	394	0.1509
seamless_m4t+whisper_v3+sb+cohere	4	whisper_v3	2611	234	0.0896
seamless_m4t+whisper_v3+sb+cohere	4	sb	2611	282	0.108
seamless_m4t+whisper_v3+sb+cohere	4	cohere	2611	118	0.0452

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+canary+conformer+parakeet	4	seamless_m4t	2611	411	0.1574
seamless_m4t+canary+conformer+parakeet	4	canary	2611	174	0.0666
seamless_m4t+canary+conformer+parakeet	4	conformer	2611	272	0.1042
seamless_m4t+canary+conformer+parakeet	4	parakeet	2611	101	0.0387
seamless_m4t+canary+conformer+sb	4	seamless_m4t	2611	375	0.1436
seamless_m4t+canary+conformer+sb	4	canary	2611	215	0.0823
seamless_m4t+canary+conformer+sb	4	conformer	2611	230	0.0881
seamless_m4t+canary+conformer+sb	4	sb	2611	248	0.095
seamless_m4t+canary+conformer+cohere	4	seamless_m4t	2611	400	0.1532
seamless_m4t+canary+conformer+cohere	4	canary	2611	198	0.0758
seamless_m4t+canary+conformer+cohere	4	conformer	2611	261	0.1
seamless_m4t+canary+conformer+cohere	4	cohere	2611	134	0.0513
seamless_m4t+canary+parakeet+sb	4	seamless_m4t	2611	411	0.1574
seamless_m4t+canary+parakeet+sb	4	canary	2611	169	0.0647
seamless_m4t+canary+parakeet+sb	4	parakeet	2611	106	0.0406
seamless_m4t+canary+parakeet+sb	4	sb	2611	304	0.1164
seamless_m4t+canary+parakeet+cohere	4	seamless_m4t	2611	427	0.1635
seamless_m4t+canary+parakeet+cohere	4	canary	2611	164	0.0628
seamless_m4t+canary+parakeet+cohere	4	parakeet	2611	109	0.0417
seamless_m4t+canary+parakeet+cohere	4	cohere	2611	150	0.0574

subpool_ranks_librispeech_validated_all_voting_ensemble (first 60 of 1158 rows)

pool	k	model	rank	count
seamless_m4t+whisper_v3+canary+conformer	4	seamless_m4t	6	841
seamless_m4t+whisper_v3+canary+conformer	4	seamless_m4t	7	159
seamless_m4t+whisper_v3+canary+conformer	4	whisper_v3	4	8
seamless_m4t+whisper_v3+canary+conformer	4	whisper_v3	5	9
seamless_m4t+whisper_v3+canary+conformer	4	whisper_v3	6	142
seamless_m4t+whisper_v3+canary+conformer	4	whisper_v3	7	841
seamless_m4t+whisper_v3+canary+conformer	4	canary	3	1000
seamless_m4t+whisper_v3+canary+conformer	4	conformer	4	719
seamless_m4t+whisper_v3+canary+conformer	4	conformer	5	279
seamless_m4t+whisper_v3+canary+conformer	4	conformer	6	2
seamless_m4t+whisper_v3+canary+conformer	4	parakeet	1	895
seamless_m4t+whisper_v3+canary+conformer	4	parakeet	2	105
seamless_m4t+whisper_v3+canary+conformer	4	sb	4	273
seamless_m4t+whisper_v3+canary+conformer	4	sb	5	712
seamless_m4t+whisper_v3+canary+conformer	4	sb	6	15

pool	k	model	rank	count
seamless_m4t+whisper_v3+canary+conformer	4	cohere	1	105
seamless_m4t+whisper_v3+canary+conformer	4	cohere	2	895
seamless_m4t+whisper_v3+canary+parakeet	4	seamless_m4t	5	8
seamless_m4t+whisper_v3+canary+parakeet	4	seamless_m4t	6	688
seamless_m4t+whisper_v3+canary+parakeet	4	seamless_m4t	7	304
seamless_m4t+whisper_v3+canary+parakeet	4	whisper_v3	4	1
seamless_m4t+whisper_v3+canary+parakeet	4	whisper_v3	5	60
seamless_m4t+whisper_v3+canary+parakeet	4	whisper_v3	6	244
seamless_m4t+whisper_v3+canary+parakeet	4	whisper_v3	7	695
seamless_m4t+whisper_v3+canary+parakeet	4	canary	1	4
seamless_m4t+whisper_v3+canary+parakeet	4	canary	2	55
seamless_m4t+whisper_v3+canary+parakeet	4	canary	3	941
seamless_m4t+whisper_v3+canary+parakeet	4	conformer	4	999
seamless_m4t+whisper_v3+canary+parakeet	4	conformer	5	1
seamless_m4t+whisper_v3+canary+parakeet	4	parakeet	1	403
seamless_m4t+whisper_v3+canary+parakeet	4	parakeet	2	577
seamless_m4t+whisper_v3+canary+parakeet	4	parakeet	3	20
seamless_m4t+whisper_v3+canary+parakeet	4	sb	5	931
seamless_m4t+whisper_v3+canary+parakeet	4	sb	6	68
seamless_m4t+whisper_v3+canary+parakeet	4	sb	7	1
seamless_m4t+whisper_v3+canary+parakeet	4	cohere	1	593
seamless_m4t+whisper_v3+canary+parakeet	4	cohere	2	368
seamless_m4t+whisper_v3+canary+parakeet	4	cohere	3	39
seamless_m4t+whisper_v3+canary+sb	4	seamless_m4t	5	357
seamless_m4t+whisper_v3+canary+sb	4	seamless_m4t	6	524
seamless_m4t+whisper_v3+canary+sb	4	seamless_m4t	7	119
seamless_m4t+whisper_v3+canary+sb	4	whisper_v3	5	132
seamless_m4t+whisper_v3+canary+sb	4	whisper_v3	6	74
seamless_m4t+whisper_v3+canary+sb	4	whisper_v3	7	794
seamless_m4t+whisper_v3+canary+sb	4	canary	3	326
seamless_m4t+whisper_v3+canary+sb	4	canary	4	674
seamless_m4t+whisper_v3+canary+sb	4	conformer	3	674
seamless_m4t+whisper_v3+canary+sb	4	conformer	4	326
seamless_m4t+whisper_v3+canary+sb	4	parakeet	1	735
seamless_m4t+whisper_v3+canary+sb	4	parakeet	2	265
seamless_m4t+whisper_v3+canary+sb	4	sb	5	511
seamless_m4t+whisper_v3+canary+sb	4	sb	6	402
seamless_m4t+whisper_v3+canary+sb	4	sb	7	87
seamless_m4t+whisper_v3+canary+sb	4	cohere	1	265
seamless_m4t+whisper_v3+canary+sb	4	cohere	2	735

pool	k	model	rank	count
seamless_m4t+whisper_v3+canary+cohere	4	seamless_m4t	5	9
seamless_m4t+whisper_v3+canary+cohere	4	seamless_m4t	6	672
seamless_m4t+whisper_v3+canary+cohere	4	seamless_m4t	7	319
seamless_m4t+whisper_v3+canary+cohere	4	whisper_v3	4	3
seamless_m4t+whisper_v3+canary+cohere	4	whisper_v3	5	58